Practice 6 + Practice 7

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Group: Wednesday, 12-14

Lab: Design and Implement a VLSM Addressing Scheme

192.168.33.128/25
255.255.255.10000000
192.168.33.10000000
10000000

network address 192.161.33.128
broadcast address 192.168.33.255

1) how many host addresses available
32-25=7 $2^7-2=126$

2) what is the total number of host address needed in the topology diagram?

BR1 LAN: 40 hosts
BR1-BR2 link: 2 hosts
BR2 LAN: 25 hosts
BR2 IOT LAN (future): 5 hosts
BR2 CCTV LAN (future): 4 hosts
BR2 HVAC LAN (future): 4 hosts

Overall: 80 hosts

3) how many subnets are needed in the network topology?

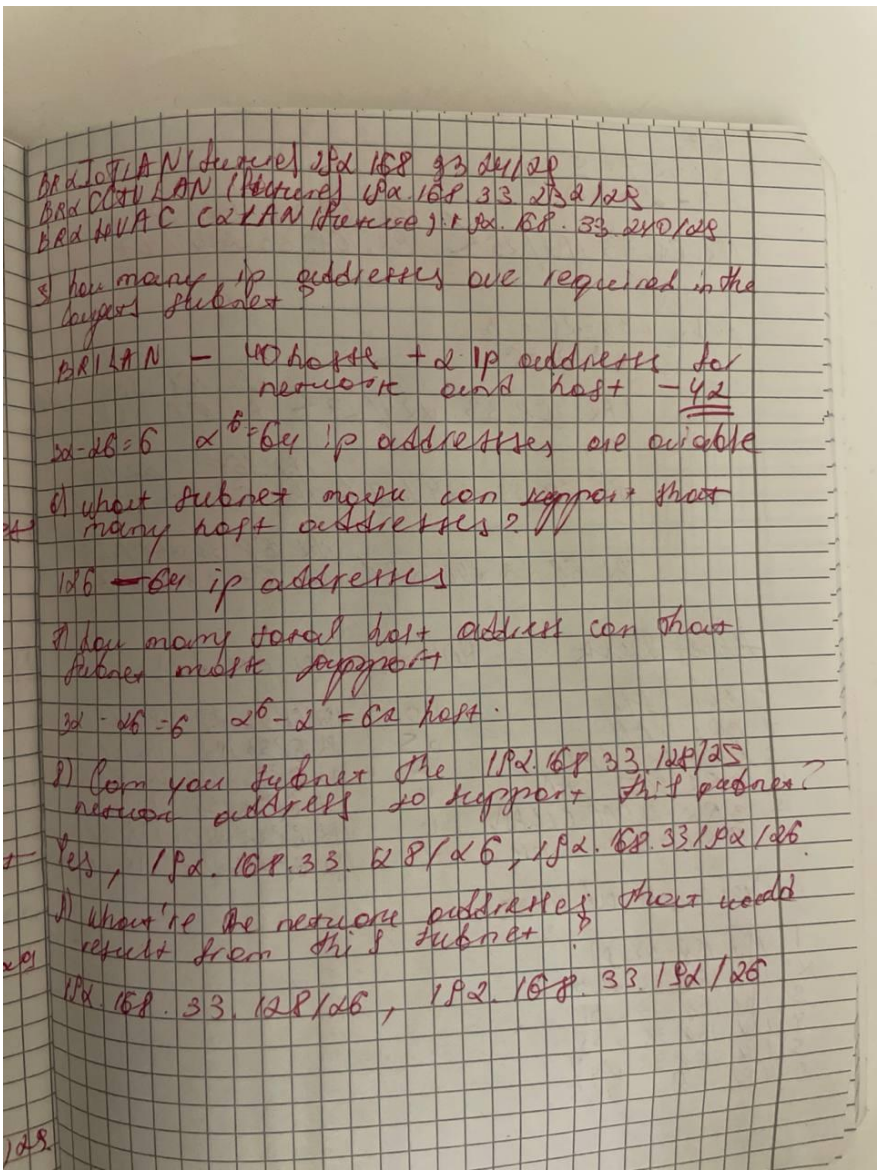
5 192.168.33.240/28

40 – 25 – 5 – 4 – 2

1 192.168.33.128/26 3 192.168.33.224/28 4 192.168.33.224/29
2 192.168.33.192/27 5 192.168.33.232/29

6 subnets are needed in the network topology

4) subnets: BR1 LAN: 192.168.33.128/26
BR1-BR2 link: 192.168.33.224/29
BR2 LAN: 192.168.33.192/27


BR2 IoT LAN (Future) 192.168.33.240/28
BR2 CCTV LAN (Future) 192.168.33.232/28
BR2 HVAC CZ LAN (Future) 192.168.33.240/28

1) how many ip addresses are required in the largest subnet?

BR1 LAN – 40 hosts + 2 IP addresses for network and host – 42

$2^6 - 2^6 = 6$ $2^6 = 64$ ip addresses are available

2) what subnet mask can support that many host addresses?

128 – 64 ip addresses

3) how many total host addresses can that subnet mask support

$2^7 - 2^6 = 6$ $2^6 - 2 = 62$ host.

4) Can you subnet the 192.168.33.128/25 network address to support this number?

Yes, 192.168.33.128/26, 192.168.33.192/26

5) what're the network addresses that would result from this subnet?

192.168.33.128/26, 192.168.33.192/26

Step 3: Determine the second largest subnet

What is the subnet description? BR2 LAN

How many IP addresses are required for the second largest subnet? 25 Hosts and 2 for Network and Broadcast IP address. 27 in total.

What subnet mask can support that many host addresses? /27

How many total host addresses can that subnet mask support and What are the network addresses that would result from this subnetting?

Can you subnet the remaining subnet again and still support this subnet? No, the number of IP addresses will not be enough.

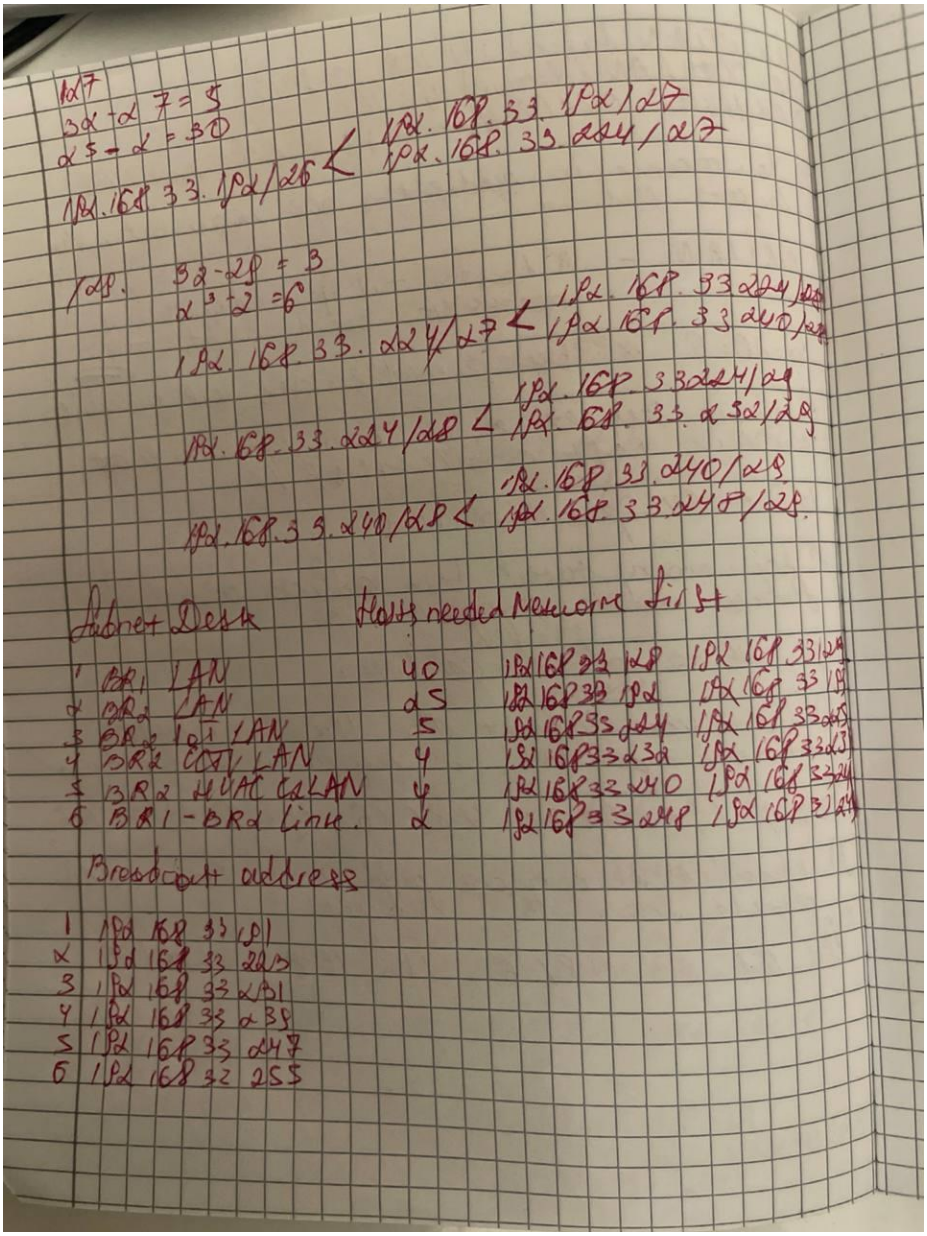
Step 4: Determine the third largest subnet.

What is the subnet description? BR2 IoT LAN

How many IP addresses are required for the next largest subnet? 5 Hosts and 2 for Network and Broadcast IP address. 7 in total.

What subnet mask can support that many host addresses? /29

How many total host addresses can that subnet mask support and What are the network addresses that would result from this subnetting?



Step 5: Determine the fourth largest subnet.

What is the subnet description? BR2 CCTV LAN and BR2 HVAC C2LAN

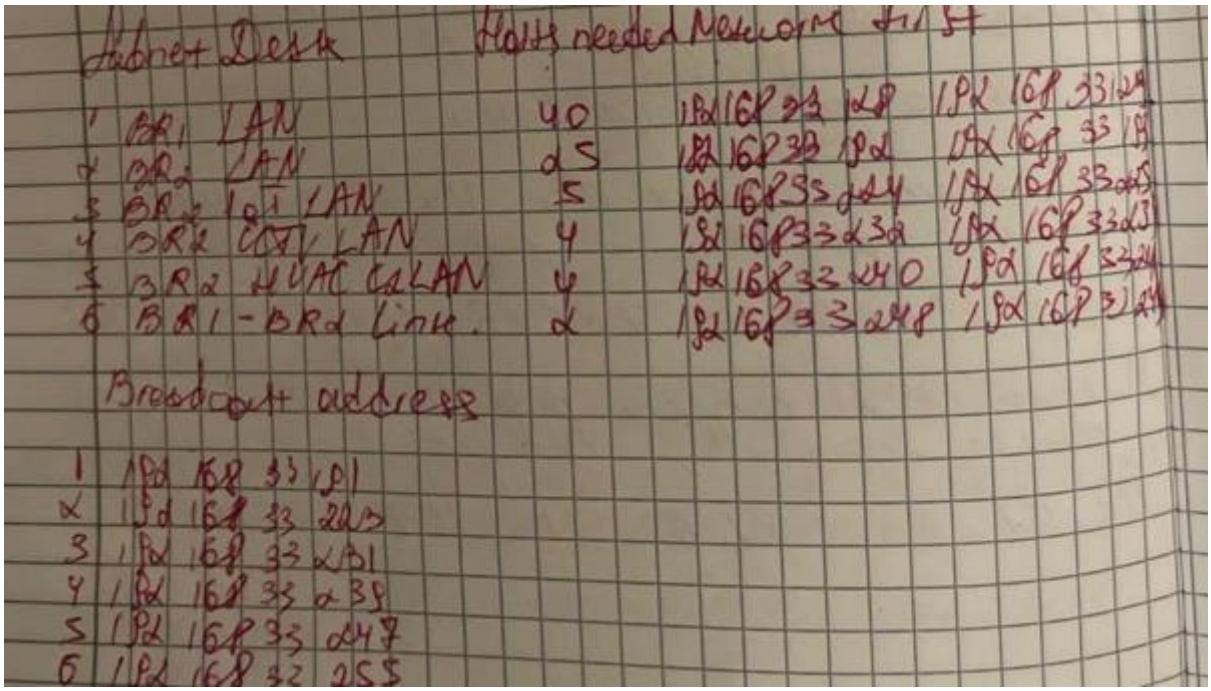
How many IP addresses are required for the next largest subnet? 4 Hosts and 2 for Network and Broadcast IP address. 6 in total.

What subnet mask can support that many host addresses? /29

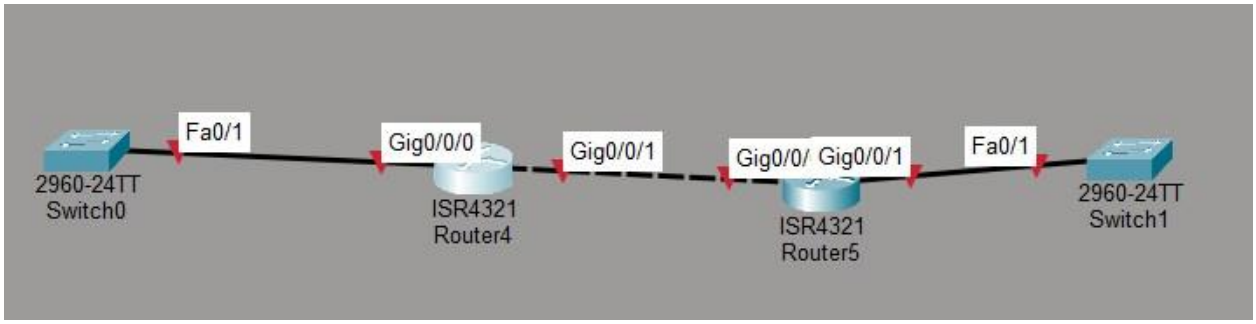
How many total host addresses can that subnet mask support and What are the network addresses that would result from this subnetting?

Part 2: Design the VLSM Address Scheme

Step 1:



Part 3: Cable and Configure the IPv4 Network



Step 2: Configure basic settings on each router.

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname BR1
BR1(config)#
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname BR2
```

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname BR1
BR1(config)#no ip domain-lookup
BR1(config)#
```

```
BR2(config)#no ip domain-lookup
```

Assign class as the privileged EXEC encrypted password for both routers.

Assign cisco as the console password and enable login for the routers.

Assign cisco as the VTY password and enable login for the routers.

Encrypt the plaintext passwords for the routers.

Create a banner that will warn anyone accessing the device that unauthorized access is prohibited on both routers.

```
BR1(config)#enable secret class
BR1(config)#line console 0
BR1(config-line)#password cisco
BR1(config-line)#login
BR1(config-line)#line vty 0 4
BR1(config-line)#password cisco
BR1(config-line)#login
BR1(config-line)#service password-encryption
BR1(config)#service password-encryption
BR1(config)#banner motd #Unauthorized access is prohibited#
BR1(config)#
```

```
BR2(config)#enable secret class
BR2(config)#line console 0
BR2(config-line)#password cisco
BR2(config-line)#login
BR2(config-line)#line vty 0 4
BR2(config-line)#password cisco
BR2(config-line)#login
BR2(config-line)#exit
BR2(config)#service password-encryption
BR2(config)#banner motd #Unauthorized access#
BR2(config)#
```

Step 3: Configure the interfaces on each router.

```
BR1(config-if)#interface GigabitEthernet0/0/0
BR1(config-if)#ip address 192.168.33.249 255.255.255.252
BR1(config-if)#no shut

BR1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

BR1(config-if)#
```

```
BR1(config)#interface GigabitEthernet0/0/1
BR1(config-if)#ip address 192.168.33.129 255.255.255.192
BR1(config-if)#ip address 192.168.33.129 255.255.255.192
BR1(config-if)#no shut

BR1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

BR1(config-if)#
```

```
BR2(config)#interface GigabitEthernet0/0/0
BR2(config-if)#ip address 192.168.33.250 255.255.255.252
BR2(config-if)#ip address 192.168.33.250 255.255.255.252
BR2(config-if)#
BR2(config-if)#exit
BR2(config)#interface GigabitEthernet0/0/1
BR2(config-if)#ip address 192.168.33.193 255.255.255.224
BR2(config-if)#ip address 192.168.33.193 255.255.255.224
```

Configure an interface description for each interface.

```
BR1(config-if)#int g0/0/0
BR1(config-if)#desc BR1-BR2 Link
BR1(config-if)#int g0/0/1
BR1(config-if)#desc BR1 LAN
```

```
BR2(config-if)#int g0/0/0
BR2(config-if)#desc BR1-BR2 Link
BR2(config-if)#int g0/0/1
BR2(config-if)#desc BR2 LAN
```

To activate the interfaces no shutdown

Step 4: Save the configuration on all devices.

copy running-config start-config

Step 5: Test Connectivity.

a. From BR1, ping BR2's G0/0/0 interface.

b. From BR2, ping BR1's G0/0/0 interface.

c. Troubleshoot connectivity issues if pings were not successful.

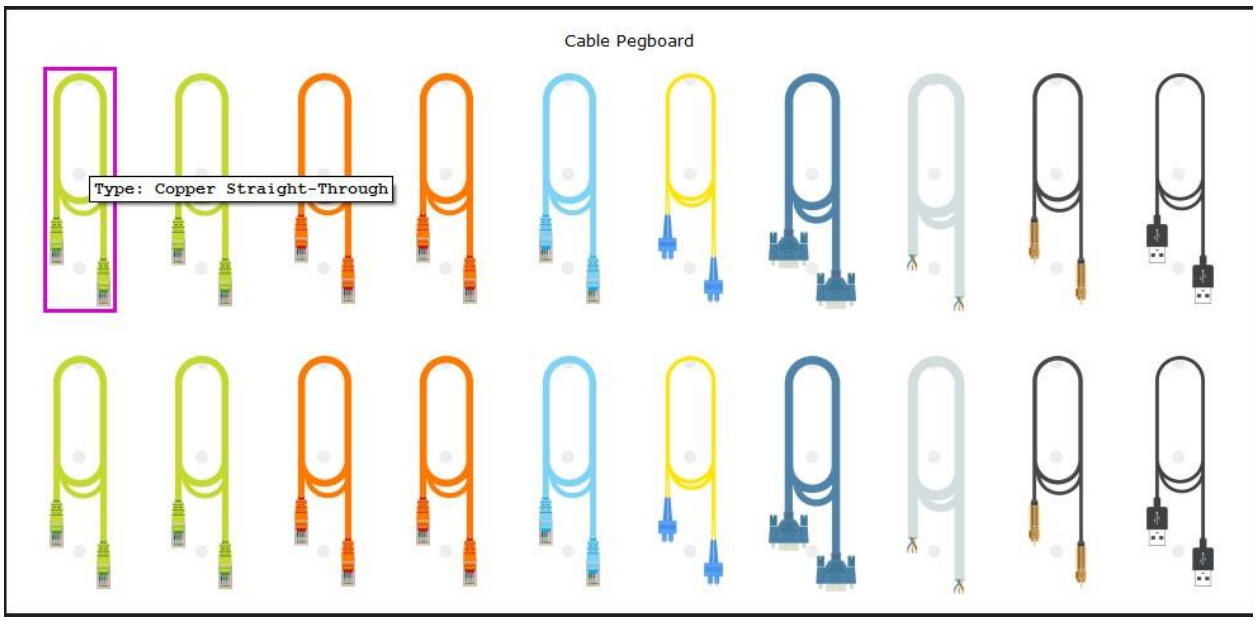
```
BR1#ping 192.168.33.250

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.33.250, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

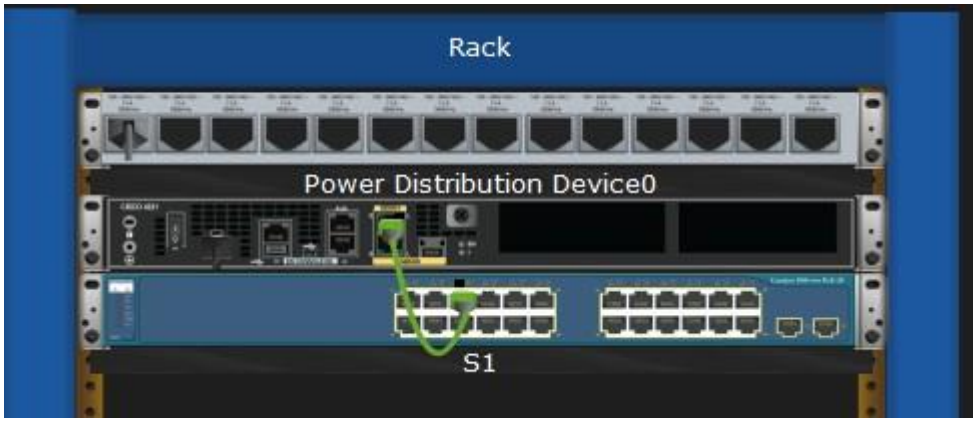
```
BR2#ping 192.168.33.249

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.33.249, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

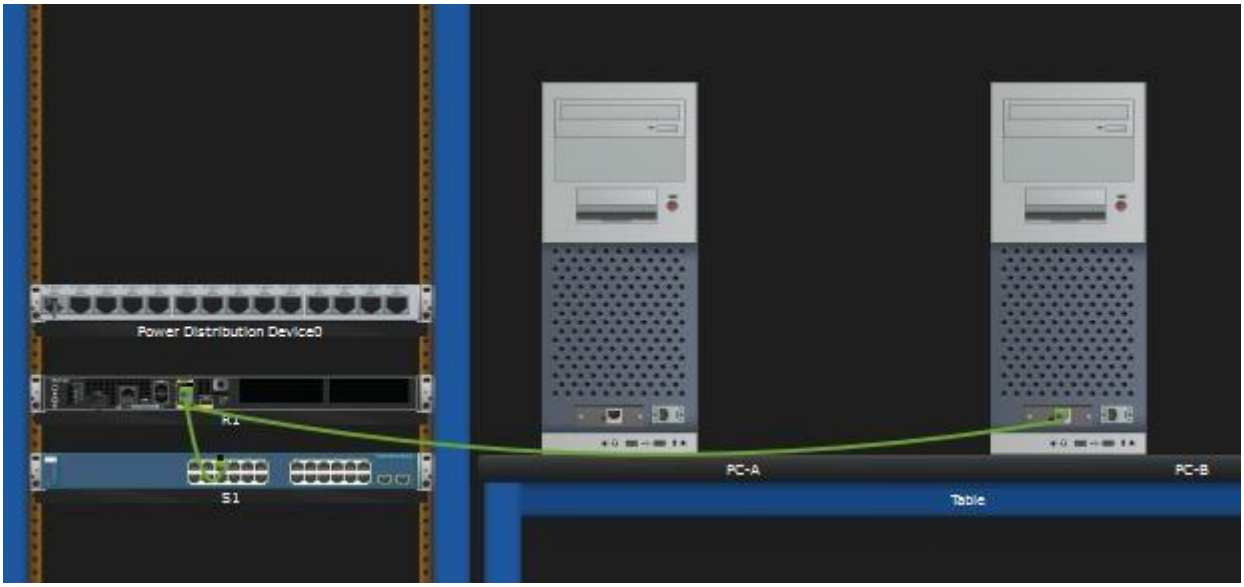
Can you think of a shortcut for calculating the network addresses of consecutive /30 subnets?

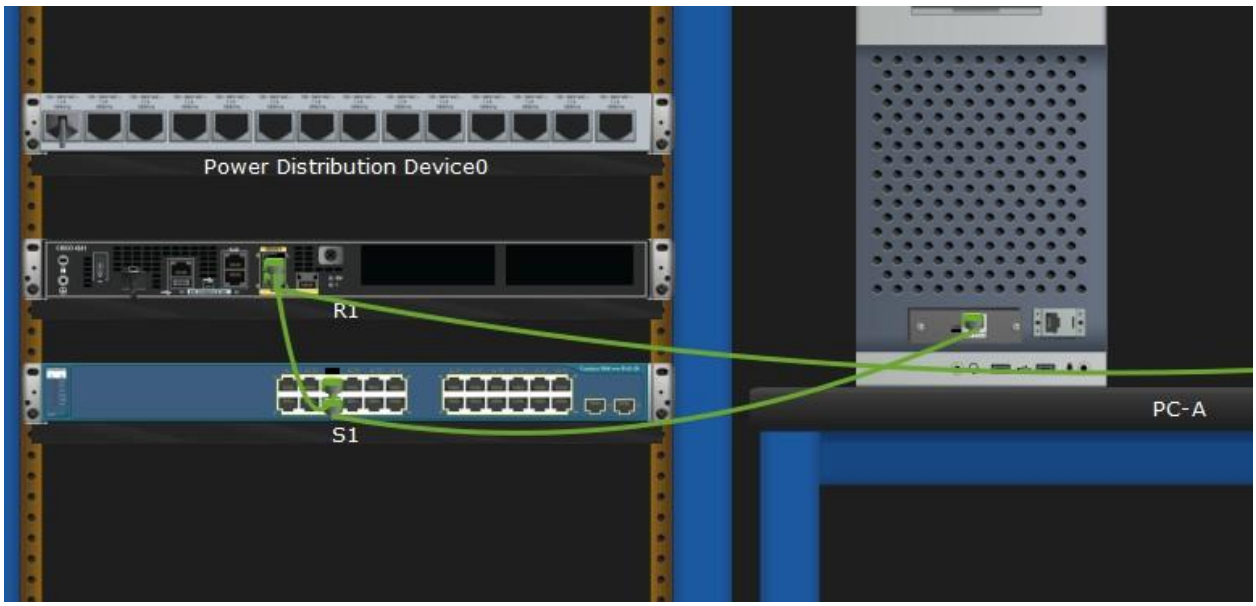
For subnets with a /30 mask, the number of bits allocated to hosts is 32 - 30 = 2. This yields 2^2 = 4 addresses in each subnet, including the network address and broadcast address. Consequently, the network addresses of consecutive /30 subnets differ by 4. To get the address of the next subnet, simply add 4 to the network address of the previous subnet. Example: the first subnet, 192.168.1.0/30, includes addresses from 192.168.1.0 to 192.168.1.3, the next subnet, 192.168.1.4/30, includes addresses from 192.168.1.4 to 192.168.1.7, and so on.

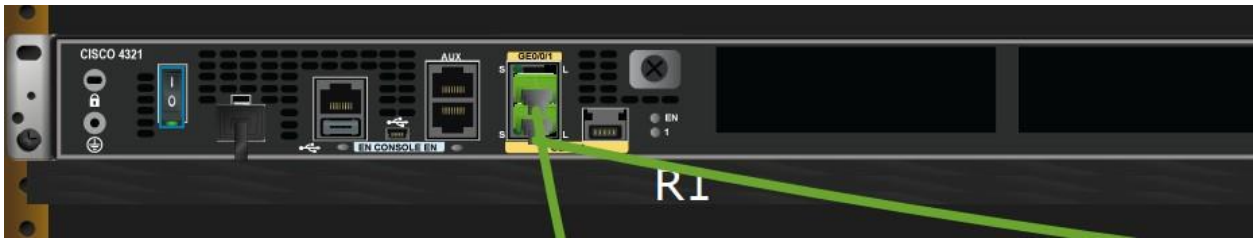
Packet Tracer: Configure IPv6 Addresses on Network Devices

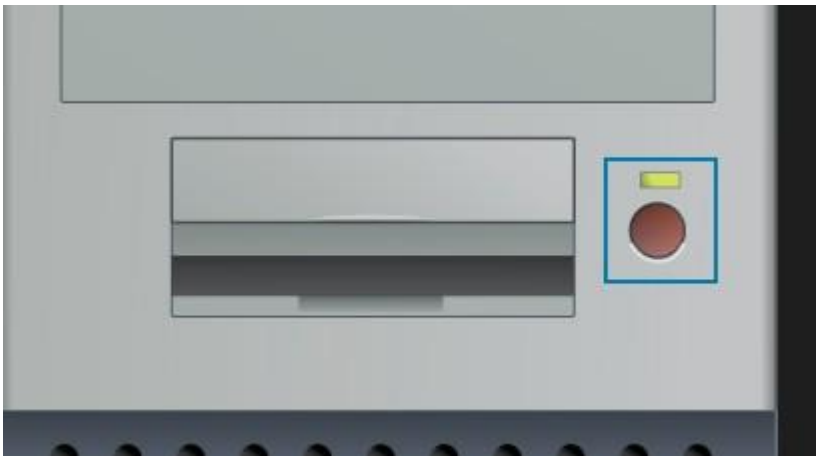

Cable Pegboard
Type: Copper Straight-Through


Rack
Power Distribution Device0
S1


Power Distribution Device0
R1
S1
PC-A
PC-B
Table

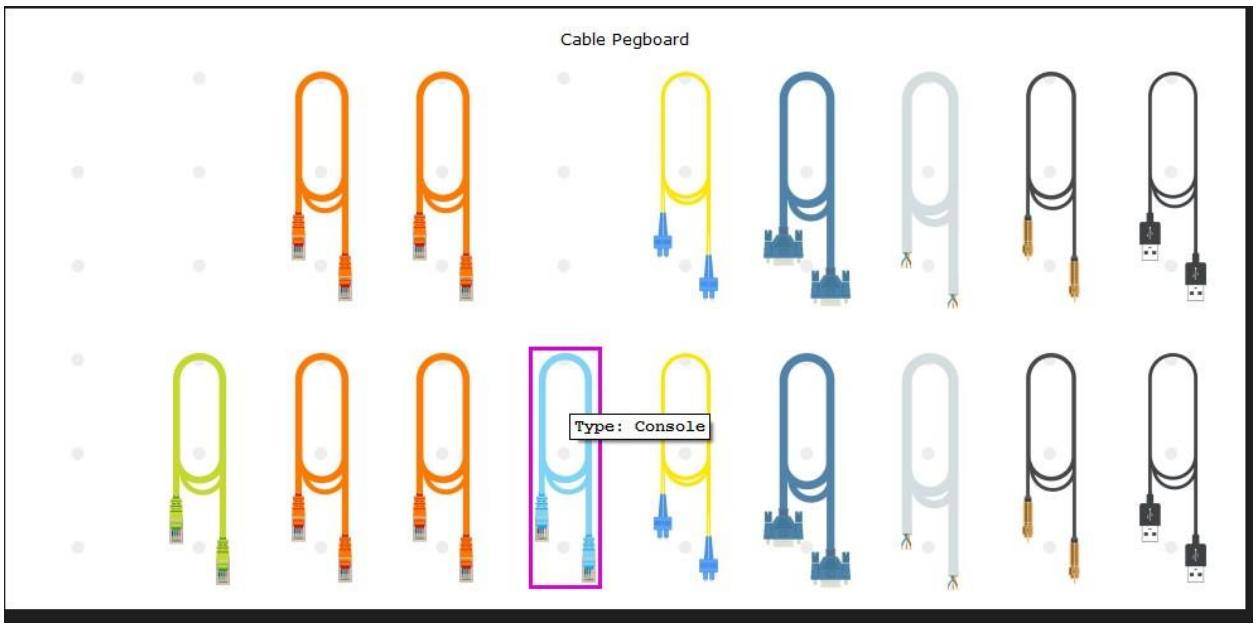


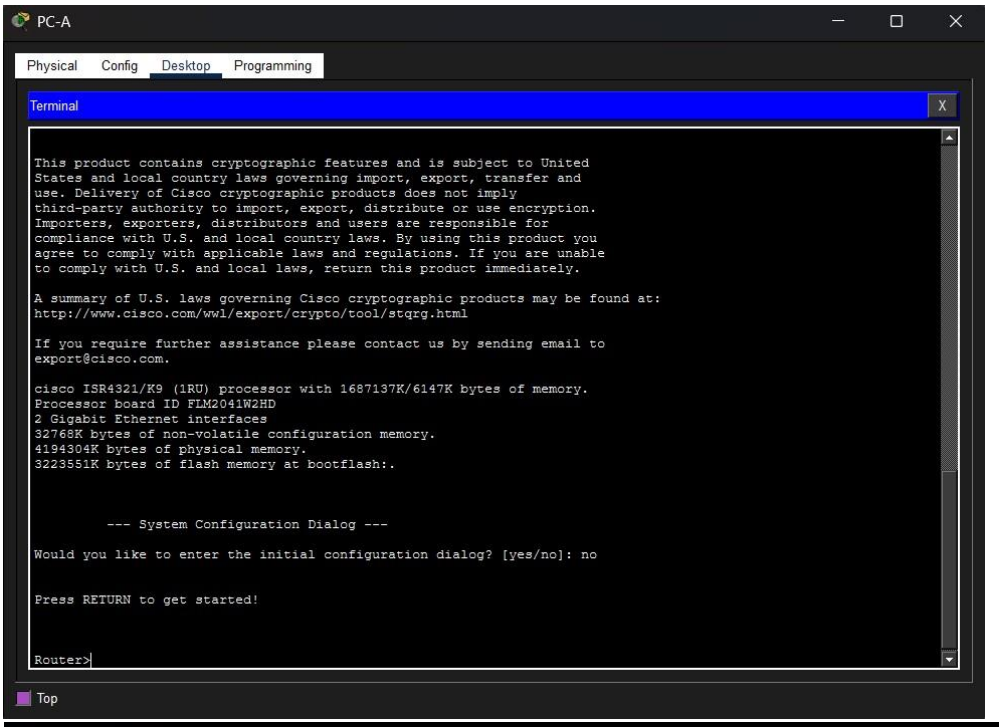




Part 2. Configure the router

To configure the router we have to choose console cable and connect the Router(Console) to PC-B(RS232)

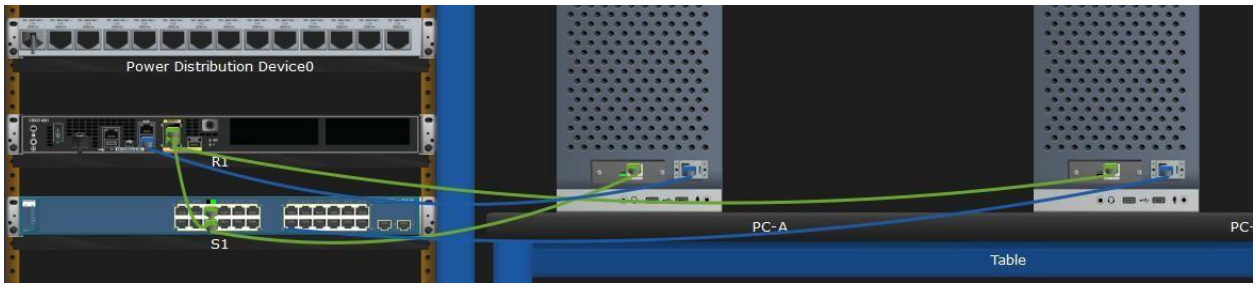





```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Step 3: Configure the switch

To configure the switch we have to choose console cable and connect the Router(Console) to PC-B(RS232). In order to find console port we have to right click on Switch and choose Inspect Rear. Then we can find console port



click to PC-B -> Desktop -> Terminal -> OK

Part 2: Configure IPv6 Addresses Manually

Step 1: Assign the IPv6 addresses to Ethernet interfaces on R1

```
R1(config-if)#int g0/0/0
R1(config-if)#ipv6 address 2001:db8:acad:a::1
% Incomplete command.
R1(config-if)#ipv6 address 2001:db8:acad:a::1/64
R1(config-if)#int g0/0/1
R1(config-if)#ipv6 address 2001:db8:acad:1::1/64
R1(config-if)#
```

```
R1#show ipv6 interface brief
GigabitEthernet0/0/0       [up/up]
    FE80::201:63FF:FE06:C001
    2001:DB8:ACAD::1
    2001:DB8:ACAD:A::1
GigabitEthernet0/0/1       [up/up]
    FE80::201:63FF:FE06:C002
    2001:DB8:ACAD:1::1
Vlan1                      [administratively down/down]
    unassigned
R1#
```

```
R1(config)#int g0/0/0
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#int g0/0/1
R1(config-if)#ipv6 address fe80::1 link-local
R1(config-if)#
```

```
R1#show ipv6 interface brief
GigabitEthernet0/0/0       [up/up]
    FE80::1
    2001:DB8:ACAD::1
    2001:DB8:ACAD:A::1
GigabitEthernet0/0/1       [up/up]
    FE80::1
    2001:DB8:ACAD:1::1
Vlan1                      [administratively down/down]
    unassigned
```

Which two multicast groups have been assigned to interface G0/0/0? FF02::1

FF02::1:FF00:1

```
R1#show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::1
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:ACAD::1, subnet is 2001:DB8:ACAD::/64
    2001:DB8:ACAD:A::1, subnet is 2001:DB8:ACAD:A::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds
```

Step 2: Enable IPv6 routing on R1.

a. From a PC-B command prompt, enter the ipconfig command to examine the IPv6 address information assigned to the PC interface.
   Question: Has an IPv6 unicast address been assigned to the network interface card (NIC) on PC-B? No, only Link-local Ipv6 Address

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:58FF:FE51:5E24
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

```
R1(config)#ipv6 unicast-routing
```

Use a command to verify the new multicast group is assigned to interface G0/0/0.

```
R1#show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::1
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:ACAD::1, subnet is 2001:DB8:ACAD::/64
    2001:DB8:ACAD:A::1, subnet is 2001:DB8:ACAD:A::/64
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:1
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
```

PC-B automatically received the Global Routing Prefix and Subnet ID through the SLAAC (Stateless Address Autoconfiguration) process. Once IPv6 unicast routing was enabled, R1 began sending Router Advertisement (RA) messages. These messages contain the prefix information, allowing the host to dynamically configure its own GUA (Global Unicast Address) without a DHCPv6 server.

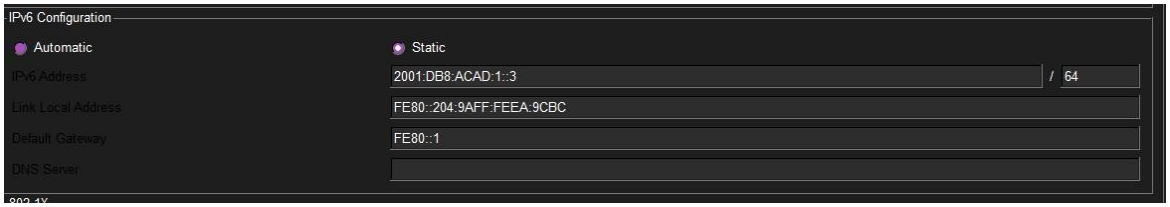
```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:58FF:FE51:5E24
   IPv6 Address....................: 2001:DB8:ACAD:A:2D0:58FF:FE51:5E24
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: FE80::1
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

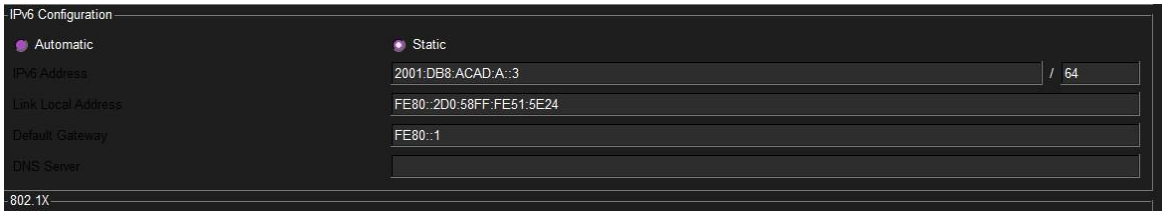
Step 3: Assign IPv6 addresses to the management interface (SVI) on S1.

```
S1(config)#int vlan 1
S1(config-if)#ipv6 address 2001:db8:acad:1::b/6
S1(config-if)#ipv6 address fe80::b link-local
S1(config-if)#no shut
```

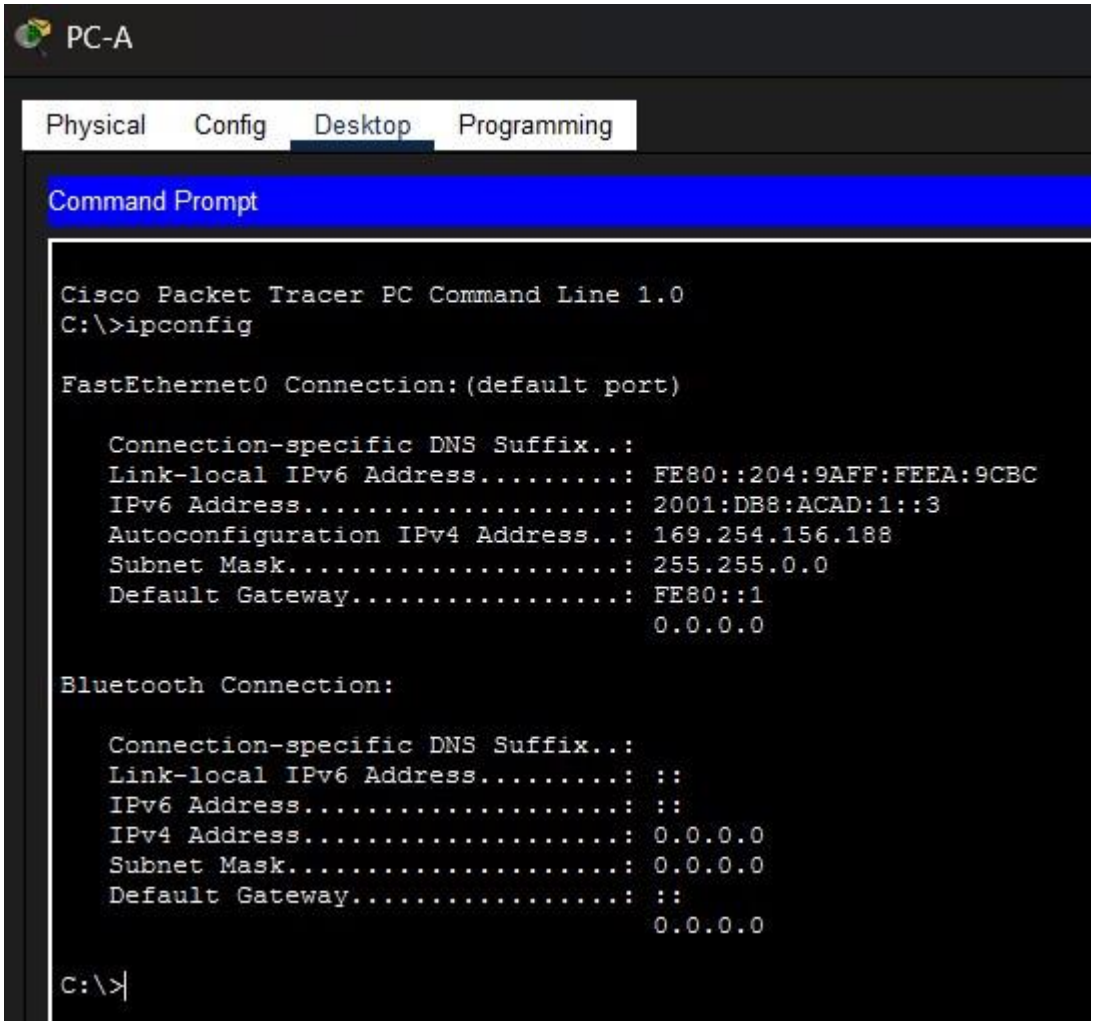
```
S1#show ipv6 interface vlan 1
Vlan1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::B
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8:ACAD:1::B, subnet is 2001:DB8:ACAD:1::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:B
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  Output features: Check hwidb
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds
S1#
```

Step 4: Assign static IPv6 addresses to the PCs.





Verify both PCs have the correct IPv6 address information.



PC-B

Physical Config Desktop Programming

Command Prompt

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:58FF:FE51:5E24
   IPv6 Address....................: 2001:DB8:ACAD:A::3
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: FE80::1
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
```

Part 3: Verify End-to-End Connectivity

From PC-A, ping fe80::1

```
C:\>ping fe80::1

Pinging fe80::1 with 32 bytes of data:

Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255

Ping statistics for FE80::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

From PC-A

```
C:\>tracert 2001:db8:acad:a::3

Tracing route to 2001:db8:acad:a::3 over a maximum of 30 hops:

  1   0 ms      0 ms      4 ms      2001:DB8:ACAD:1::1
  2   0 ms      3 ms      0 ms      2001:DB8:ACAD:A::3

Trace complete.
```

From PC-B, ping PC-A

```
C:\>ping 2001:db8:acad:1::3

Pinging 2001:db8:acad:1::3 with 32 bytes of data:

Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:1::3: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:1::3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping fe80::1

Pinging fe80::1 with 32 bytes of data:

Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255
Reply from FE80::1: bytes=32 time<1ms TTL=255

Ping statistics for FE80::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Reflection Questions

1. Why can the same link-local address, fe80::1, be assigned to both Ethernet interfaces on R1?

Link-local addresses (FE80::/10) have a scope limited strictly to a single physical or logical link. Since each Ethernet interface on R1 represents a distinct network segment, using the same address (fe80::1) does not cause a conflict. The router differentiates between these identical addresses by associating them with their specific exit interface (Index/Zone ID).

2. What is the Subnet ID of the IPv6 unicast address 2001:db8:acad::aaaa:1234/64, if the global routing prefix is a /48?

In a /64 address with a /48 global routing prefix, the Subnet ID is contained within the 4th hextet (bits 49 to 64).

- Address: 2001:0db8:acad:0000:aaaa:1234/64
- The 4th hextet is 0000.

Therefore, the Subnet ID is 0